Table 1: Validity (%), joint MMD, and reference coverage (%) across five DGPs (three-seed means). Bold marks the best displayed method for each DGP and metric.

Method	Weighted Discrete			Linear			Cubic			Smoothstep			2D Bridge		
	Valid. ↑	J-MMD ↓	Cover. ↑	Valid. ↑	J-MMD ↓	Cover. ↑	Valid. ↑	J-MMD ↓	Cover. ↑	Valid. ↑	J-MMD ↓	Cover. ↑	Valid. ↑	J-MMD ↓	Cover. ↑
CD	13.55	0.1467	40.00	83.22	0.0025	78.52	78.12	0.0062	74.35	85.55	0.0034	70.49	80.65	0.0072	73.08
Compositional CD	75.50	0.1065	84.78	38.54	0.0139	31.25	31.74	0.0368	27.67	21.21	0.0497	16.06	20.93	0.1054	20.04
Additive mean†	0.00	0.1174	0.00	**100.00**	0.0088	0.55	0.00	0.1711	0.00	0.00	0.0716	0.00	0.00	0.3416	0.00
Additive + residual†	0.18	0.2322	0.65	7.81	0.1407	6.17	0.00	0.2037	0.00	1.35	0.1523	0.86	0.18	0.3266	0.10
Fixed composition + residual†	0.13	0.2353	0.50	0.03	0.2327	0.00	0.02	0.2014	0.00	0.07	0.2548	0.00	0.00	0.3798	0.00
Rep4Ex-inspired + residual†	0.85	0.2714	2.87	3.68	0.2490	3.06	0.23	0.2732	0.02	0.39	0.2621	0.02	0.03	0.3648	0.03
CG: sparse factorized ($s = 1$)	46.93	0.1027	74.85	5.40	0.0776	1.84	8.92	0.1280	4.07	3.43	0.1096	0.75	4.07	0.2045	3.52
CG: dense factorized ($s = 1$)	55.43	0.1073	75.58	7.67	0.0692	2.86	3.45	0.1266	0.65	2.86	0.1073	0.80	1.69	0.2084	1.79
CG: dense joint ($s = 1$)	1.47	0.5295	4.27	15.35	0.0359	6.74	13.22	0.0745	5.49	14.75	0.0526	4.57	6.18	0.1360	4.82
CG: sparse factorized ($s = 2$)	76.13	0.0530	61.22	24.82	0.0161	7.42	35.74	0.0183	13.85	19.48	0.0383	4.87	36.04	0.0758	30.99
CG: dense factorized ($s = 2$)	**82.85**	0.0633	56.77	29.61	0.0116	9.83	26.16	0.0194	2.67	17.01	0.0423	1.14	5.70	0.0752	4.64
CG: dense joint ($s = 2$)	0.45	0.5508	1.42	18.20	0.0111	9.18	34.49	0.0143	13.77	25.72	0.0241	6.35	14.19	0.0394	9.52
SEDGE: no weighting	12.00	0.2198	29.95	8.98	0.4006	20.82	8.97	0.3650	17.61	8.92	0.4248	17.12	2.46	0.6294	4.10
SEDGE-LGD ($s = 0.5$)	38.07	0.1525	70.00	44.11	0.1343	64.81	31.92	0.2048	49.61	47.90	0.1267	60.01	23.67	0.3420	35.40
SEDGE-LGD ($s = 1$)	56.80	0.0996	81.08	78.24	0.0104	79.17	60.17	0.0711	64.49	79.57	0.0147	70.05	66.37	0.1153	66.60
SEDGE-LGD ($s = 2$)	39.98	0.0272	57.12	77.07	0.0302	62.42	71.89	0.0302	50.91	79.46	0.0328	55.27	76.25	0.0516	56.77
SEDGE-LWS	77.55	**0.0225**	**89.97**	91.98	**0.0018**	**87.37**	**91.26**	**0.0024**	**84.29**	**91.55**	**0.0030**	**82.23**	**87.39**	**0.0016**	**78.86**

Validity requires both specifications and root support. J-MMD compares joint (X, Z) distributions. Coverage is the fraction of reference samples covered under the original fixed-radius criterion. Higher validity/coverage and lower J-MMD are preferred; compare methods within each DGP. †Paper-inspired adaptations, not faithful reproductions. Additive mean is a deterministic control rather than a posterior sampler. Existing LGD scales are all shown; CG scales 1 and 2 are shown. The expanded LGD sweep is not included. Weighted Discrete is a relaxed Assumption-3 control (conditional exclusion 2.5% and 5%); continuous settings meet the working 1% criterion. This is a results-informed selection, not a claim of uniform superiority. No significance claim is implied by boldface.